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Date

Week-2 Module

1. Fraud Transaction Detection (Linear Search)

Problem: A bank stores transaction IDs in order they occurs. Since the transaction are not sorted, the security team needs to check if a suspicious transaction ID exists.

Code:

```
import java.util.Scanner;
public class LinearSearch {
    public static void main (String[] args) {
        Scanner s = new Scanner (System.in);
        int n = s.nextInt();
        int [] arr = new int[n];
        for (int i=0; i<n; i++) {
            arr[i] = s.nextInt();
        }
        int sus = s.nextInt();
        boolean found = false;
        for (int i=0; i<n; i++) {
            if (arr[i] == sus) {
                found = true;
                break;
            }
        }
```

Date

```
if (found) {
    System.out.println("Fraud Transaction
                        ID found");
} else {
    System.out.println("Transaction not found");
}
s.close();
  }
}
```

LinearSearch.java

Share Run

```java
import java.util.Scanner;

public class LinearSearch {
    public static void main(String[] args) {

        Scanner s = new Scanner(System.in);

        int n = s.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = s.nextInt();
        }

        int sus = s.nextInt();

        boolean found = false;

        for(int i = 0; i < n; i++) {
            if(arr[i] == sus) {
                found = true;
                break;
            }
        }

        if(found) {
            System.out.println("Fraud Transaction Found");
        } else {
            System.out.println("Transaction Not Found");
        }

        s.close();
    }
}
```

Output

```
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found

=== Code Execution Successful ===
```

2. Server Log Search (Binary Search)

Problem: A cloud company stores server log IDs in sorted order. Engineers need to quickly verify if a log ID exists.

Code:

```
import java.util.Scanner;
public class BinarySearch {
    public static void main(String[] args){
        Scanner s = new Scanner(System.in);
        int n= s.nextInt();
        int[] arr = new int[n];
        for(int i=0; i<n; i++){
            arr[i]= s.nextInt();
        }
        int target = s.nextInt();
        int start=0;
        int end = n-1;
        boolean found = false;
```

Date

```
while (start <= end){
    int mid = start + (end-start)/2;
    if (arr[mid] == target){
        found = true;
        break;
    }
    else if (arr[mid] < target){
        start = mid + 1;
    }
    else {
        end = mid - 1;
    }
}
if (found) {
    System.out.println("Log found");
}else {
    System.out.println("Log Not found");
}
s.close();
    }
}
```

BinarySearch.java

Share Run

```java
import java.util.Scanner;

public class BinarySearch {
    public static void main(String[] args) {

        Scanner s = new Scanner(System.in);
        int n = s.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = s.nextInt();
        }

        int target = s.nextInt();
        int start = 0;
        int end = n - 1;
        boolean found = false;

        while(start <= end) {
            int mid = start + (end - start) / 2;
            if(arr[mid] == target) {
                found = true;
                break;
            }
            else if(arr[mid] < target) {
                start = mid + 1;
            }
            else {
                end = mid - 1;
            }
        }
        if(found) {
            System.out.println("Log Found");
        } else {
            System.out.println("Log Not Found");
        }
        s.close();
    }
}
```

Output

```
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found

=== Code Execution Successful ===
```

3. Sort Employee Salaries (Bubble Sort)

Problem : A company wants to ~~st~~ sort employee salaries in ascending order for payroll analysis.

Code :

```
import java.util.Scanner;
public class BubbleSort {
    public static void main(String [] args) {
        Scanner s = new Scanner(System.in);
        int n= s.nextInt();
        int [] arr = new int [n];
        for ( int i=0; i<n; i++){
            arr[i] = s.nextInt();
        }
        for (int i=0; i<n-1; i++) {
            for (int j =0; j<n-i-1; j++){
                if ( arr[j] > arr[j+1]) {
                    int temp = arr[j] ;
                    arr[j] = arr[j+1];
                    arr[j+1]= temp;
                }
            }
        }
        for (int n : arr) {
            System.out.print(n + " ");
        }
        s.close();
    }
}
```

BubbleSort.java

Share

Run

```java
import java.util.Scanner;

public class BubbleSort {
    public static void main(String[] args) {

        Scanner s = new Scanner(System.in);

        int n = s.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = s.nextInt();
        }

        for(int i = 0; i < n - 1; i++) {

            for(int j = 0; j < n - i - 1; j++) {

                if(arr[j] > arr[j + 1]) {

                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }

        for(int num : arr) {
            System.out.print(num + " ");
        }

        s.close();
    }
}
```

Output

```
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
=== Code Execution Successful ===
```